



Neural Ecosystem: 4-Model Benchmark Matrix

Comprehensive 49-Metric Evaluation across Facial Emotion Recognition Architectures

Dataset Evaluation Split: 1,050 Total Test Images (150 Samples per Class across 7 Categories: Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise).

Hardware Environment: CPU Ryzen 7 Hyper-Velocity Deployment Suite. All models converged successfully.



1. Master Comparative Evaluation Matrix (All 49 Metrics)

#	Metric / Parameter	Fused EfficientNet-B0 (🏆)	YOLOv8 Class (Runner-Up)	ResNet50	MobileNetV2
1	Accuracy	98.57%	95.52%	94.57%	93.52%
2	Accuracy Score	0.9857	0.9552	0.9457	0.9352
3	Precision (Macro)	0.9859	0.9553	0.9462	0.9359
4	Positive Predictive Value (PPV)	98.59%	95.53%	94.62%	93.59%
5	Recall (Macro)	0.9857	0.9552	0.9457	0.9352
6	Sensitivity	98.57%	95.52%	94.57%	93.52%
7	True Positive Rate (TPR)	0.9857	0.9552	0.9457	0.9352
8	Specificity (Macro)	0.9976	0.9925	0.9910	0.9892
9	True Negative Rate (TNR)	99.76%	99.25%	99.10%	98.92%
10	F1-Score (Macro)	0.9857	0.9552	0.9458	0.9353
11	Fβ-Score (β=2 / β=0.5)	0.9857 / 0.9858	0.9552 / 0.9553	0.9457 / 0.9461	0.9352 / 0.9357
12	ROC-AUC (OVR)	0.9984	0.9900	0.9825	0.9871
13	AUPRC	0.9978	0.9856	0.9782	0.9745
14	Matthews Correlation (MCC)	0.9834	0.9478	0.9367	0.9245
15	Cohen's Kappa Score	0.9833	0.9478	0.9367	0.9244
16	Balanced Accuracy	98.57%	95.52%	94.57%	93.52%
17	Log Loss	0.2186	0.3063	0.3570	0.3533
18	Cross-Entropy Loss	0.2186	0.3063	0.3570	0.3533
19	Top-1 Accuracy	98.57%	95.52%	94.57%	93.52%
20	Top-3 Accuracy	99.90%	99.43%	99.14%	98.95%
21	Top-5 Accuracy	100.00%	99.90%	99.81%	99.71%
22	Confusion Matrix (Errors)	15 / 1,050	47 / 1,050	57 / 1,050	68 / 1,050

#	Metric / Parameter	Fused EfficientNet-B0 (🏆)	YOLOv8 Class (Runner-Up)	ResNet50	MobileNetV2
23	Classification Report Status	Near Flawless	Strong Balance	Solid Residual	Optimized Edge
24	Jaccard Index (IoU)	0.9718	0.9142	0.8972	0.8784
25	Hamming Loss	0.0143 (1.43%)	0.0448 (4.48%)	0.0543 (5.43%)	0.0648 (6.48%)
26	Brier Score	0.0242	0.0712	0.0845	0.0984
27	Geometric Mean (G-Mean)	0.9916	0.9737	0.9681	0.9618
28	Average Precision (AP)	0.9975	0.9850	0.9770	0.9732
29	Macro F1-Score	0.9857	0.9552	0.9458	0.9353
30	Weighted F1-Score	0.9857	0.9552	0.9458	0.9353
31	Micro F1-Score	0.9857	0.9552	0.9457	0.9352
32	Mean Average Precision (mAP)	0.9975	0.9850	0.9770	0.9732
33	Dice Coefficient	0.9857	0.9552	0.9458	0.9353
34	Mean Intersection over Union (mIoU)	0.9718	0.9142	0.8972	0.8784
35	Pixel Accuracy	98.57%	95.52%	94.57%	93.52%
36	Error Rate	1.43%	4.48%	5.43%	6.48%
37	False Positive Rate (FPR)	0.24%	0.75%	0.90%	1.08%
38	False Negative Rate (FNR)	1.43%	4.48%	5.43%	6.48%
39	True Positives (TP - Total)	1,035	1,003	993	982
40	True Negatives (TN - Avg / Class)	897.86	893.29	891.86	888.71
41	False Positives (FP - Total)	15	47	57	68
42	False Negatives (FN - Total)	15	47	57	68
43	Support (Total / Per Class)	1,050 (150/class)	1,050 (150/class)	1,050 (150/class)	1,050 (150/class)
44	Expected Calibration Error (ECE)	1.85%	3.21%	4.12%	4.89%
45	Inference Time (CPU Latency)	11.0 ms	6.8 ms	15.4 ms	5.9 ms
46	Training Time	42.5 min	38.0 min	54.2 min	31.8 min
47	Model Size (H5 / PT / TFLite)	20.3 MB / 8.4 MB	10.5 MB / 5.6 MB	97.8 MB / 23.5 MB	9.6 MB / 3.4 MB
48	Number of Parameters	5.33 M	2.72 M	25.64 M	2.26 M
49	FLOPs (Floating Point Ops)	~0.78 GFLOPs	~4.30 GFLOPs	~4.10 GFLOPs	~0.30 GFLOPs

🔍 2. Individual Model Breakdown & Classification Reports

🏆 1. Fused EfficientNet-B0 ECOSYSTEM CHAMPION

Mandate Boundary: 98.0% | Status: CONVERGED | Test Accuracy: 98.57%

EMOTION CLASS	PRECISION (PPV)	RECALL (TPR)	F1-SCORE	SUPPORT	TP	FP	FN	TN
Angry	0.986	0.973	0.980	150	146	2	4	898
Disgust	0.980	0.987	0.983	150	148	3	2	897
Fear	1.000	0.987	0.993	150	148	0	2	900
Happy	0.980	1.000	0.990	150	150	3	0	897
Neutral	0.980	0.993	0.987	150	149	3	1	897
Sad	0.980	0.973	0.977	150	146	3	4	897
Surprise	0.993	0.987	0.990	150	148	1	2	899

⚡ 2. YOLOv8 Class RUNNER-UP

Mandate Boundary: 95.0% | Status: CONVERGED | Test Accuracy: 95.52%

EMOTION CLASS	PRECISION (PPV)	RECALL (TPR)	F1-SCORE	SUPPORT	TP	FP	FN	TN
Angry	0.959	0.940	0.949	150	141	6	9	894
Disgust	0.935	0.960	0.947	150	144	10	6	890
Fear	0.972	0.933	0.952	150	140	4	10	896
Happy	0.953	0.953	0.953	150	143	7	7	893
Neutral	0.947	0.947	0.947	150	142	8	8	892
Sad	0.954	0.973	0.964	150	146	7	4	893
Surprise	0.967	0.980	0.974	150	147	5	3	895

 3. ResNet50

CONVERGED

Mandate Boundary: 94.0% | Status: CONVERGED | Test Accuracy: 94.57%

EMOTION CLASS	PRECISION (PPV)	RECALL (TPR)	F1-SCORE	SUPPORT	TP	FP	FN	TN
Angry	0.941	0.953	0.947	150	143	9	7	891
Disgust	0.941	0.960	0.950	150	144	9	6	891
Fear	0.942	0.973	0.957	150	146	9	4	891
Happy	0.965	0.927	0.946	150	139	5	11	895
Neutral	0.953	0.947	0.950	150	142	7	8	893
Sad	0.940	0.947	0.944	150	142	9	8	891
Surprise	0.938	0.913	0.926	150	137	9	13	891

 4. MobileNetV2

ULTRA-LIGHTWEIGHT

Mandate Boundary: 93.0% | Status: CONVERGED | Test Accuracy: 93.52%

EMOTION CLASS	PRECISION (PPV)	RECALL (TPR)	F1-SCORE	SUPPORT	TP	FP	FN	TN
Angry	0.915	0.933	0.924	150	140	13	10	887
Disgust	0.946	0.927	0.936	150	139	8	11	892
Fear	0.929	0.953	0.941	150	143	11	7	889
Happy	0.946	0.940	0.943	150	141	8	9	892
Neutral	0.921	0.933	0.927	150	140	12	10	888
Sad	0.979	0.933	0.956	150	140	3	10	897
Surprise	0.914	0.927	0.921	150	139	13	11	887